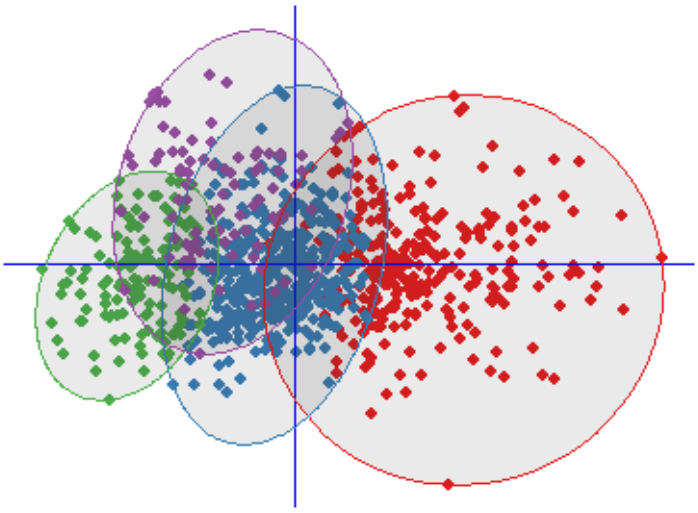
Statistical Learning

Gennaro Basile

March 26, 2026



Contents

1 Sampling Distributions & The Central Limit Theorem **3**
1.1 The Three Fundamental Distributions 3
1.2 Chebyshev's Inequality: The Safety Net 3
1.3 Convergence Laws: Why *Big Data* Works 3
1.4 The Central Limit Theorem (CLT) 4
1.5 Sum of Independent Random Variables 5

1 Sampling Distributions & The Central Limit Theorem

1.1 The Three Fundamental Distributions

In Data Science, you will constantly calculate metrics (like the mean) from a subset of data. The ultimate goal is to understand if the mean of your *sample* accurately reflects the reality of the entire *population*.

To do this without confusion, we must strictly separate three concepts:

1. Population Distribution (*The Hidden Truth*)

The abstract, true probability distribution of the entire population. In the real world, you almost never have access to this. It has a true mean (μ) and true variance (σ^2).

2. Sample Data Distribution (*Your DataFrame*)

The actual, physical data you collected (e.g., $n = 25$ restaurant bills). You can plot this as a histogram. It has a sample mean ($\bar{y}$) and sample standard deviation (s). As your dataset grows ($n \to \infty$), this histogram will start to look exactly like the Population Distribution.

3. Sampling Distribution (*The Theoretical Metric*)

This is the hardest concept to grasp. Imagine you collect $1,000$ different samples of 25 bills, and calculate the mean for *each* sample. If you plot those $1,000$ means on a new histogram, you get the **Sampling Distribution of the Mean**. It tells you how much your sample mean is expected to fluctuate simply due to random luck.

The Standard Error (SE): The standard deviation of this new Sampling Distribution is called the Standard Error:

$$SE(\bar{Y}) = \frac{\sigma}{\sqrt{n}} \tag{1}$$

Data Science Takeaway: Notice the $\sqrt{n}$ in the denominator. To cut your error margin in half, you don't need twice as much data; you need **four times** as much data.

1.2 Chebyshev's Inequality: The Safety Net

In the real world, data is rarely perfectly Normal (Gaussian). What if your data distribution is completely unknown or highly skewed?

Chebyshev's Inequality

For *any* random variable X with a known mean μ and variance σ^2, the probability that a value falls within k standard deviations from the mean is at least:

$$P(|X - \mu| < k\sigma) \geq 1 - \frac{1}{k^2} \tag{2}$$

Why it matters: If $k = 2$, $1 - (1/4) = 0.75$. This guarantees that *no matter how weird or skewed your data is*, at least 75% of your observations will always fall within 2 standard deviations of the mean. It is the ultimate statistical worst-case scenario bound.

1.3 Convergence Laws: Why *Big Data* Works

As your sample size (n) approaches infinity, uncertainty vanishes.

- **Convergence in Distribution ($\xrightarrow{d}$):** The *shape* of your sample's distribution starts to match the target distribution. (This is the basis of the Central Limit Theorem).
- **Convergence in Probability ($\xrightarrow{p}$):** The probability that your sample estimate is wrong shrinks to exactly zero.

The Law of Large Numbers (LLN)

- **Weak LLN:** As n grows, the probability that your sample mean $\bar{X}$ differs from the true mean μ by any tiny amount (ε) approaches zero.
- **Strong LLN:** The sample mean *will* converge to the true mean with a probability of exactly 1.

1.4 The Central Limit Theorem (CLT)

This is arguably the most important theorem in all of statistics and machine learning.

Central Limit Theorem

Take *any* population distribution (it can be heavily skewed, bimodal, or uniform). If you draw large enough samples ($n \geq 30$) and calculate their means, the distribution of those sample means will form a **perfect Normal (Gaussian) Distribution**.

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \tag{3}$$

Data Science Takeaway: The CLT is the reason why the Normal distribution is the default assumption for so many Machine Learning algorithms (like Linear Regression residuals). Because many real-world phenomena are the sum of thousands of tiny, independent random events, the resulting data naturally converges to a Gaussian shape.

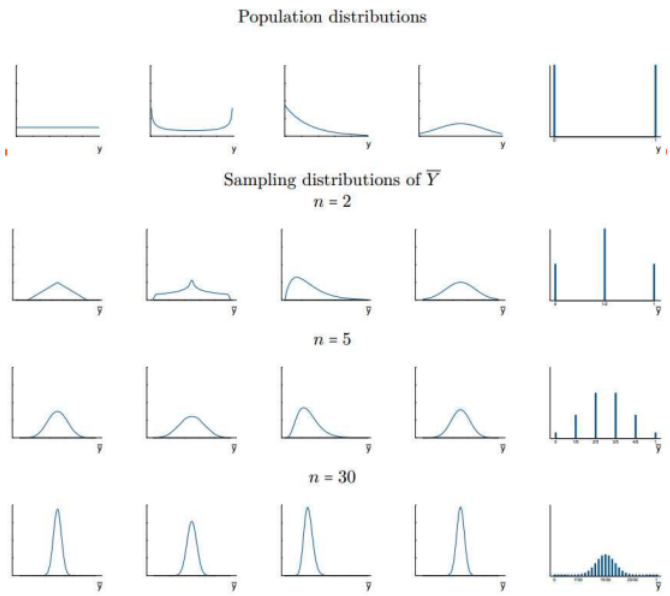


Figure 1.4.1: *Central Limit Theorem: Regardless of the starting population shape (top row), the sampling distribution of the mean becomes perfectly Normal as sample size n increases (bottom row).*

1.5 Sum of Independent Random Variables

In Machine Learning, we constantly aggregate features or build ensembles (like summing the outputs of multiple decision trees). What happens when you add probability distributions together?

Closure Under Convolution

- **Normal + Normal = Normal:** If you add two independent Normal variables, the result is still perfectly Normal. (The means add up, and the variances add up).
- **Binomial + Binomial = Binomial:** Only if they share the exact same success probability π.
- **Poisson + Poisson = Poisson:** The rate parameters (λ) simply add together.